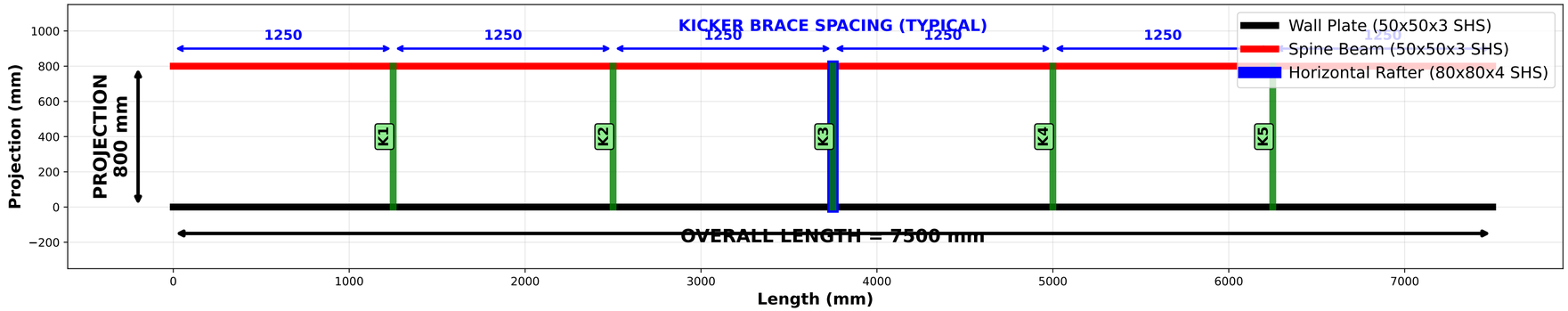
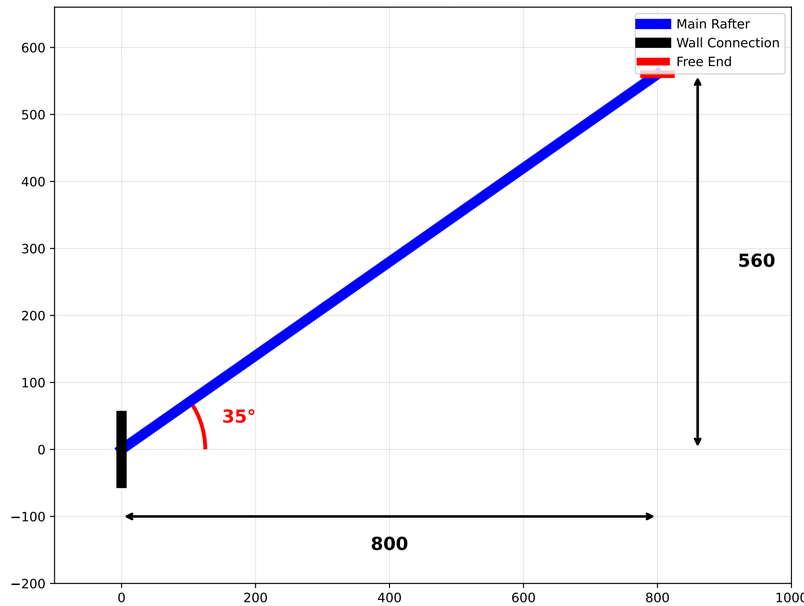
CANTILEVERED MONOPITCH AWNING - FINAL DESIGN
Steel Frame Structure

PLAN VIEW - STRUCTURAL LAYOUT



SIDE ELEVATION



SECTION DETAILS

50x50x3 SHS

80x80x4 SHS

MATERIAL LIST & SPECIFICATIONS

Member	Size	Length(mm)	Qty	Total(m)	Weight(kg)
Wall Plate	50x50x3 SHS	7500	1	7.500	34.3
Spine Beam	50x50x3 SHS	7500	1	7.500	34.3
Horizontal Rafter	80x80x4 SHS	977	1	0.977	9.4
Kicker Braces	50x50x3 SHS	977	5	4.883	22.3

TOTAL LENGTH: 20.860 m
TOTAL WEIGHT: 100.3 kg

FABRICATION NOTES & CONNECTION DETAILS

GENERAL NOTES:
• All structural steel to comply with AS/NZS 3679.1 Grade 300
• All welding to be carried out in accordance with AS/NZS 1554.1
• Hot dip galvanize all steelwork after fabrication to AS/NZS 4680
• Prime and paint with 2-pack polyurethane system

WALL CONNECTION:
• Fix wall plate to masonry/concrete wall using M16 Grade 8.8 expansion anchors
• Maximum spacing 400mm centers, minimum edge distance 100mm
• Provide 8mm thick backing plate where required

WELDING REQUIREMENTS:
• All connections to have 6mm minimum fillet welds
• Main rafter connections to be full strength butt welds
• Back gouge and re-weld critical connections
• NDT testing if specified by engineer

TOLERANCES:
• Overall length: ±5mm
• Member lengths: ±2mm
• Squareness: ±3mm per meter
• Straightness: L/500 maximum deviation